Recall that the definite integral $\int_a^b f(x)dx$ of an integrable function f on $[a,b]$ is defined by

$$\int_a^b f(x)dx = \underline{\int_a^b} f(x)dx = \overline{\int_a^b} f(x)dx,$$

where

$$\underline{\int_a^b} f(x)dx := \sup\{L(f,P) : P \text{ is a partition of } [a,b]\},$$
$$\overline{\int_a^b} f(x)dx := \inf\{U(f,P) : P \text{ is a partition of } [a,b]\}.$$

We now prove the following

Theorem. Suppose f is integrable on $[a,b]$. Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that if P is a partition of $[a,b]$ with $|P| := \max\{x_i - x_{i-1}\} < \delta$ and $\tau = \{x_i^*\}$ any tagging of P, then

$$\left| R(f,P,\tau) - \int_a^b f(x)dx \right| < \varepsilon.$$

Proof. Suppose $f : [a,b] \to \mathbb{R}$ is integrable and let $\varepsilon > 0$. Since $\int_a^b f(x)dx$ is the least upper bound of the set of all lower sums, $\int_a^b f(x)dx - \frac{\varepsilon}{2}$ is not an upper bound for this set, so there exists a partition P_1 such that

$$\int_a^b f(x)dx - \frac{\varepsilon}{2} < L(f,P_1)$$

and therefore

$$\int_a^b f(x)dx - L(f,P_1) < \frac{\varepsilon}{2}.$$

Since $L(f,P) \le \int_a^b f(x)dx$ for every partition P, we have

$$0 \le \int_a^b f(x)dx - L(f,P_1) < \frac{\varepsilon}{2}. \tag{1}$$

ID #1:_____
ID #2:_____

1. Show that there exists a partition P_2 of $[a,b]$ such that

$$0 \leq U(f,P_2) - \int_a^b f(x)dx < \frac{\varepsilon}{2}. \tag{2}$$

2. Recall that a partition P' of $[a,b]$ is a *refinement* of P of $[a,b]$ if $P \subseteq P'$, and that the Refinement Lemma says that

$$L(f,P) \leq L(f,P') \leq U(f,P') \leq U(f,P).$$

Use the Refinement Lemma to show that there exists a partition P_ε such that

$$0 \leq \int_a^b f(x)dx - L(f,P_\varepsilon) < \frac{\varepsilon}{2} \quad \text{and} \quad 0 \leq U(f,P_\varepsilon) - \int_a^b f(x)dx < \frac{\varepsilon}{2}. \tag{3}$$

ID #1:_____
ID #2:_____

3. Show that, for any partition P of $[a,b]$, and any tagging τ of P,

$$L(f,P) \leq R(f,P,\tau) \leq U(f,P).$$

4. We have seen that $L(f,P) \leq R(f,P,\tau) \leq U(f,P)$ and $L(f,P) \leq \int_a^b f(x)dx \leq U(f,P)$ for any partition P of $[a,b]$. Show that these imply $\left|R(f,P,\tau) - \int_a^b f(x)dx\right| \leq U(f,P) - L(f,P)$.

Since f is bounded on $[a,b]$, there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in [a,b]$. Let N be the number of subintervals of the partition P_ε. Choose $\delta > 0$ such that $2MN\delta < \frac{\varepsilon}{2}$.

Claim: For any partition P with $|P| < \delta$, $U(f,P) - L(f,P) < \varepsilon$. Since we have just seen that $\left|R(f,P,\tau) - \int_a^b f(x)dx\right| \leq U(f,P) - L(f,P)$, so the proof of the theorem then follows from this claim. Thus, all that remains is to prove this claim.

To prove the claim, let P be any partition of $[a,b]$ with $|P| < \delta$, and put $P' = P \cup P_\varepsilon$. Since

$$L(f,P) \leq \int_a^b f(x)dx \leq U(f,P),$$

we have

$$0 \leq \int_a^b f(x)dx - L(f,P) = \left(\int_a^b f(x)dx - L(f,P')\right) + (L(f,P') - L(f,P)), \tag{4}$$

and

$$0 \leq U(f,P) - \int_a^b f(x)dx = \left(U(f,P') - \int_a^b f(x)dx\right) + (U(f,P) - U(f,P')). \tag{5}$$

We will show that both terms on the right-hand side of each equation are both $< \frac{\varepsilon}{2}$, from which the claim follows since

$$\int_a^b f(x)dx - L(f,P) < \frac{\varepsilon}{2} \quad \text{and} \quad U(f,P) - \int_a^b f(x)dx < \frac{\varepsilon}{2}$$

imply that $U(f,P) - L(f,P) = \int_a^b f(x)dx - L(f,P) + U(f,P) - \int_a^b f(x)dx < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

5. Apply the Refinement Lemma to $P_\varepsilon \subseteq P'$ to show that

$$\int_a^b f(x)dx - L(f,P') < \frac{\varepsilon}{2} \quad \text{and} \quad U(f,P') - \int_a^b f(x)dx < \frac{\varepsilon}{2}.$$

This gives two of the four inequalities.

It remains to prove that

$$L(f,P') - L(f,P) < \frac{\varepsilon}{2},$$
$$U(f,P) - L(f,P') < \frac{\varepsilon}{2}.$$

To this end, label the points of the partition P as $P = \{a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b\}$. Since P' is a refinement of P, for each $0 \le i \le n-1$, $P'_i := P' \cap [x_i, x_{i+1}]$ is a partition of $[x_i, x_{i+1}]$.

We then have

$$L(f,P') = \sum_{i=0}^{n-1} L(f,P'_i) \quad \text{and} \quad U(f,P') = \sum_{i=0}^{n-1} U(f,P'_i),$$

where, labeling the points of P'_i as $P'_i = \{x_i = x'_{i,0} < x'_{i,1} < \cdots < x'_{i,n_i-2} < x'_{i,n_i-1} = x_{i+1}\}$,

$$L(f,P'_i) := \sum_{j=0}^{n_i-2} m'_{i,j}(x'_{i,j+1} - x'_{i,j}),$$
$$U(f,P'_i) := \sum_{j=0}^{n_i-2} M'_{i,j}(x'_{i,j+1} - x'_{i,j}),$$

with

$$m'_{i,j} := \inf\{f(x) : x'_{i,j} \le x \le x'_{i,j+1}\},$$
$$M'_{i,j} := \sup\{f(x) : x'_{i,j} \le x \le x'_{i,j+1}\}.$$

By the Refinement Lemma,

$$L(f,P) \le L(f,P') \le U(f,P') \le U(f,P),$$

from which it follows that

$$\begin{aligned} 0 \le L(f,P') - L(f,P) &= \sum_{i=0}^{n-1} (L(f,P'_i) - m_i(x_{i+1} - x_i)) \\ &= \sum_{i=0}^{n-1} \left(\sum_{j=0}^{n_i-2} m'_{ij}(x'_{i,j+1} - x'_{i,j}) - m_i \sum_{j=0}^{n_i-2} (x'_{i,j+1} - x'_{i,j}) \right) \\ &= \sum_{i=0}^{n-1} \sum_{j=0}^{n_i-2} (m'_{ij} - m_i)(x'_{i,j+1} - x'_{i,j}). \end{aligned}$$

Similarly, we have

$$0 \le U(f,P) - U(f,P') = \sum_{i=0}^{n-1} (M_i(x_{i+1} - x_i) - U(f,P'_i)) = \sum_{i=0}^{n-1} \sum_{j=0}^{n_i-2} (M_i - M'_{ij})(x'_{i,j+1} - x'_{i,j}).$$

Since $m'_{ij} := \inf\{f(x) : x_i \le x'_{i,j} \le x \le x'_{i,j+1} \le x_{i+1}\}$ and $m_i := \inf\{f(x) : x_i \le x \le x_{i+1}\}$, m_i is a lower bound for the set $\{f(x) : x_i \le x'_{i,j} \le x \le x'_{i,j+1} \le x_{i+1}\}$ and therefore must be $\le$ the greatest

lower bound of this set, which is m'_{ij}. Hence, $m_i \le m'_{ij}$ and therefore $m'_{ij} - m_i \ge 0$ for all $0 \le i \le n-1$ and $0 \le j \le n_i - 1$. Similarly, $M_i - M'_{ij}$ for all $0 \le i \le n-1$ and $0 \le j \le n_i - 1$.

If $P' \cap (x_i, x_{i+1}) = \emptyset$, then $P'_i = \{x_i = x'_{i,0} < x'_{i,1} = x_{i+1}\}$, and

$$L(f, P_i) = m'_{i,0}(x'_{i,1} - x'_{i,0}) = m_i(x_{i+1} - x_i),$$

so the corresponding term $L(f, P'_i) - m_i(x_{i+1} - x_i)$ in the sum above is zero. Similarly, we have $M_i(x_{i+1} - x_i) - U(f, P'_i) = 0$. Since P' has at most $N-1$ terms not contained in P, [1] there are at most $\widetilde{N} := \min\{n, N-1\}$ indices i such that (x_i, x_{i+1}) contains at least one point of P'_i. [2] As we have just seen, for all other indices the terms are zero. Letting the index k run over the subset of indices i for which the corresponding terms are non-zero, we have

$$0 \le L(f, P') - L(f, P) = \sum_{k=0}^{\widetilde{N}-1} \sum_{j=0}^{n_k - 2} (m'_{kj} - m_k)(x'_{k,j+1} - x_{k,j}),$$

$$0 \le U(f, P) - U(f, P') = \sum_{k=0}^{\widetilde{N}-1} \sum_{j=0}^{n_k - 2} (M_k - M'_{kj})(x'_{k,j+1} - x'_{k,j}).$$

6. Show that $-M \le f(x) \le M$ for all $x \in [a, b]$ implies that

$$m'_{kj} - m_k \le 2M \quad \text{and} \quad M_k - M'_{kj} \le 2M.$$

[1] P_ε has $N+1$ points and the endpoints a, b are also in P, so this leaves $N-1$.

[2] If $n < N-1$ and none of the remaining $N-1$ points of P_ε are in P, then there is at least one point of P_ε in each of the n subintervals. If $n \ge N-1$, then at most $N-1$ subintervals contain a point of P_ε if we put one point from P_ε in each until we run out of points. Drawing a picture for a small value of n, say 3, may help to visualize this.

7. Complete the proof.

□